

#	Video	Key	PC1	Visual density (OpenCV)			Physical surprise				Behaviour	
				Mean	SD	Z	Mean ^a	Max ^a	Z	Zmax	RTonset (s)	ACC (%)
Concept Verification												
Highest PC1												
1	GapNarrow	591b1bce	18.10	1.505	0.230	-0.07	867.50	871.76	-0.75	+0.08	8.50	93.5
2	GapSplit	179cfd14	15.04	1.550	0.237	+0.04	867.43	871.44	-0.79	-0.09	8.76	92.3
3	GapTall	253669d6	10.41	1.513	0.224	-0.05	867.53	871.78	-0.73	+0.09	9.26	95.0
4	GapDoor	e27791af	10.35	1.545	0.242	+0.02	867.46	871.43	-0.77	-0.09	9.36	95.0
5	BalloonTop	937f0d80	6.51	2.006	0.079	+1.10	868.91	871.33	+0.22	-0.14	9.88	93.1
6	BasketOrange	ca9ff607	6.21	1.914	0.049	+0.89	867.81	871.51	-0.54	-0.05	9.39	91.2
7	DarkCorner	4cc16b46	5.52	1.266	0.187	-0.63	867.07	873.08	-1.04	+0.76	9.51	95.4
8	WoodBox	c308dfea	4.49	0.833	0.126	-1.64	867.97	873.39	-0.43	+0.92	11.78	88.9
9	DarkCorner	88fdec3b	4.30	1.248	0.173	-0.67	867.30	873.10	-0.88	+0.77	9.57	95.0
10	TwinCups	ebc75ed6	3.99	1.895	0.077	+0.84	868.95	869.82	+0.24	-0.92	9.77	92.0
Lowest PC1												
1	BasketGreen	d386d692	-12.61	1.915	0.048	+0.89	867.85	871.54	-0.51	-0.04	10.97	90.0
2	TriPrismClose	34bccc18	-9.18	1.415	0.268	-0.28	870.03	872.72	+0.98	+0.57	10.88	76.2
3	BalloonTop	1196c2cd	-8.42	2.005	0.080	+1.10	868.92	871.33	+0.22	-0.14	10.91	91.2
4	CupCage	e40af308	-7.81	1.873	0.078	+0.79	867.11	868.12	-1.02	-1.80	10.40	90.0
5	TriPrism	04c1ec25	-7.05	1.418	0.260	-0.27	870.43	873.06	+1.25	+0.75	10.76	75.5
6	CupMagenta	f084a168	-6.59	1.873	0.077	+0.79	867.20	868.26	-0.95	-1.73	10.44	93.5
7	TwinCups	e853c91d	-6.56	1.895	0.078	+0.84	868.87	869.71	+0.19	-0.98	10.37	89.7
8	CupCageEmpty	15660a01	-6.26	1.873	0.080	+0.79	867.08	868.13	-1.04	-1.80	10.34	91.2
9	CupCage	561c3ba7	-5.18	1.875	0.076	+0.79	867.08	868.15	-1.03	-1.79	10.31	95.0
10	CupMagenta	5872c022	-4.83	1.875	0.077	+0.79	867.20	868.28	-0.95	-1.72	9.92	95.0
Affordance Recognition												
Highest PC1												
1	GapTall	253669d6	8.51	1.513	0.224	-0.05	867.53	871.78	-0.73	+0.09	9.66	93.4
2	GapNarrow	591b1bce	8.01	1.505	0.230	-0.07	867.50	871.76	-0.75	+0.08	11.26	89.9
3	GapSplit	179cfd14	5.72	1.550	0.237	+0.04	867.43	871.44	-0.79	-0.09	11.35	87.2
4	GapDoor	e27791af	5.00	1.545	0.242	+0.02	867.46	871.43	-0.77	-0.09	10.63	85.3
5	WoodShadow	29bb6f59	4.70	0.709	0.088	-1.93	868.27	873.53	-0.22	+0.99	10.17	97.7
6	DarkCorner	0f8eb818	4.61	1.241	0.176	-0.68	867.27	873.12	-0.91	+0.78	10.80	8.5
7	WoodBox	613bda98	3.37	0.738	0.119	-1.86	868.21	873.53	-0.26	+0.99	10.93	88.4
8	YellowCube	40b790e8	3.35	0.826	0.169	-1.65	870.49	872.75	+1.29	+0.59	10.57	7.4
9	PlainPillar	7d58767f	3.24	1.061	0.484	-1.10	872.03	874.88	+2.34	+1.69	10.44	96.5
10	BlackBlock	b8ef9a57	3.20	1.220	0.333	-0.73	871.58	874.65	+2.04	+1.57	10.90	95.0
Lowest PC1												
1	DarkCorner	4cc16b46	-13.47	1.266	0.187	-0.63	867.07	873.08	-1.04	+0.76	13.29	30.2
2	BasketClose	e5b0981b	-9.92	1.913	0.047	+0.88	867.79	871.46	-0.55	-0.08	12.54	83.3
3	BalloonTop	837511d2	-9.11	2.007	0.079	+1.10	868.89	871.31	+0.20	-0.15	12.54	86.8
4	BalloonTop	1196c2cd	-7.36	2.005	0.080	+1.10	868.92	871.33	+0.22	-0.14	11.81	87.2
5	BasketOrange	ca9ff607	-6.79	1.914	0.049	+0.89	867.81	871.51	-0.54	-0.05	11.98	84.5
6	BasketGreen	19106a00	-5.35	1.915	0.048	+0.89	867.77	871.44	-0.56	-0.09	11.92	88.0
7	BalloonTop	a193e9c4	-4.29	2.000	0.083	+1.09	868.93	871.35	+0.23	-0.13	11.25	93.4
8	TriPrism	04c1ec25	-3.62	1.418	0.260	-0.27	870.43	873.06	+1.25	+0.75	12.52	75.6
9	TwinCups	65983493	-2.74	1.896	0.076	+0.84	868.88	869.73	+0.19	-0.97	11.75	87.6
10	BasketGreen	277f8147	-2.32	1.914	0.046	+0.89	867.78	871.46	-0.56	-0.07	11.42	93.0
Plausibility Assessment												
Highest PC1												
1	GapSplit	179cfd14	16.87	1.550	0.237	+0.04	867.43	871.44	-0.79	-0.09	8.54	96.3
2	GapNarrow	591b1bce	16.69	1.505	0.230	-0.07	867.50	871.76	-0.75	+0.08	8.41	96.3
3	GapTall	253669d6	10.51	1.513	0.224	-0.05	867.53	871.78	-0.73	+0.09	9.06	95.1
4	GapDoor	e27791af	8.86	1.545	0.242	+0.02	867.46	871.43	-0.77	-0.09	9.38	93.1
5	BasketOrange	ca9ff607	7.90	1.914	0.049	+0.89	867.81	871.51	-0.54	-0.05	9.23	89.0
6	TwinCups	ebc75ed6	6.13	1.895	0.077	+0.84	868.95	869.82	+0.24	-0.92	9.37	93.9
7	DarkCorner	4cc16b46	5.53	1.266	0.187	-0.63	867.07	873.08	-1.04	+0.76	9.23	94.3
8	DarkCorner	88fdec3b	5.47	1.248	0.173	-0.67	867.30	873.10	-0.88	+0.77	9.27	93.9
9	BasketClose	e5b0981b	5.16	1.913	0.047	+0.88	867.79	871.46	-0.55	-0.08	9.65	93.1
10	WoodBox	c308dfea	4.35	0.833	0.126	-1.64	867.97	873.39	-0.43	+0.92	9.32	91.1
Lowest PC1												
1	BasketGreen	d386d692	-12.24	1.915	0.048	+0.89	867.85	871.54	-0.51	-0.04	10.49	89.0
2	BalloonTop	1196c2cd	-11.64	2.005	0.080	+1.10	868.92	871.33	+0.22	-0.14	10.56	91.5
3	CupMagenta	f084a168	-8.02	1.873	0.077	+0.79	867.20	868.26	-0.95	-1.73	10.68	89.8
4	CupCage	e40af308	-7.29	1.873	0.078	+0.79	867.11	868.12	-1.02	-1.80	10.20	85.4
5	CupCageEmpty	15660a01	-6.89	1.873	0.080	+0.79	867.08	868.13	-1.04	-1.80	10.26	90.2
6	CupMagenta	bc9d04d6	-5.72	1.874	0.077	+0.79	867.12	868.22	-1.00	-1.75	10.18	90.7
7	CupMagenta	5872c022	-5.12	1.875	0.077	+0.79	867.20	868.28	-0.95	-1.72	10.15	91.9
8	TwinCups	e853c91d	-5.03	1.895	0.078	+0.84	868.87	869.71	+0.19	-0.98	9.91	89.4
9	TriPrismClose	34bccc18	-5.00	1.415	0.268	-0.28	870.03	872.72	+0.98	+0.57	10.09	78.9
10	BasketGreen	19106a00	-4.89	1.915	0.048	+0.89	867.77	871.44	-0.56	-0.09	10.19	95.1

^aRaw surprise in units of 10⁻³. PC1 from the video×participant RT_{onset} matrix, per condition.